Deep Learning for Natural Language Processing

Today's goal • 2/3 items

Module 1
Feedforward Neural Nets and Recurrent Neural Networks

Introduction to the Course
- Course Updates and Accessibility Support
  Reading • 1 min
- Earn Academic Credit for Your Work!
  Reading • 10 min
- Course Support
  Reading • 10 min
- Assessment Expectations
  Reading • 5 min
- **AI Citation and Acknowledgement**
  Reading • 10 min

Feedforward Networks
- The Perceptron
  Video • 6 min
- The XOR Problem
  Video • 4 min
- Feedforward Networks
  Video • 7 min
- Feedforward Networks in NLP
  Video • 8 min
- Mathematics of Feedforward Networks
  Reading • 25 min
- Sentence Embeddings with Feedforward Networks
  Video • 5 min
- Feedforward Networks
  Practice Assignment • 15 min

Recurrent Neural Networks
- Sentence Embeddings with Recurrent Neural Networks
  Video • 4 min
- Sequence Labelling with Recurrent Neural Networks
  Video • 4 min
- Mathematics of Recurrent Neural Networks
  Reading • 25 min
- LSTMs and GRUs
  Video • 6 min
- Bidirectional Recurrent Neural Networks
  Video • 3 min
- Hierarchical Recurrent Neural Networks
  Video • 5 min
- Recurrent Neural Networks in Python
  Video • 8 min
- Recurrent Neural Networks
  Practice Assignment • 15 min

Implementing and Training Neural Networks
- Loss Functions
  Video • 6 min
- The Intuition behind Gradient Descent
  Video • 5 min
- Stochastic Gradient Descent
  Video • 6 min
- Stochastic Gradient Descent
  Reading • 25 min
- Stochastic Gradient Descent in Python
  Video • 6 min
- Stochastic Gradient Descent
  Lab • 30 min
- Training Neural Networks
  Practice Assignment • 15 min

Review and Exercises
- AI Policy Quiz
  Graded Assignment • 5 min
- Feedforward Networks, Recurrent Neural Networks, and How to Train Them
  Graded Assignment • 15 min
- Recurrent Neural Networks for Sequence Labelling
  Programming Assignment • 1h 30m

Module 2
Sequence to Sequence Models, Attention, Transformers

Module 3
Transfer Learning

Module 4
Large Language Models

AI Citation and Acknowledgement

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For both written and coding assignments, any outputs from AI that are incorporated, including direct quotation, paraphrasing, and rephrasing, should be cited appropriately. A citation should also be given for outputs from the AI resulting from editing, grammar correction, or translation. For written assignments, use the citation style specified or used in the course, and if this is not given, please use APA. For code, put a comment above the code with the corresponding citation and the lines it applies to. Please refer to Brown University's guide to Citation and Attribution for AI tools for more details.

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